

KRISHNA PETROCHEMICALS PVT. LTD.

Visakhapatnam Refinery — Zone 4 (Crude Distillation Unit)

PUMP STATION MAINTENANCE & INSPECTION REPORT

Report No.	KPL-MNT-2026-08831	Date	22-Aug-2026
Equipment Tag	PMP-CDU4-P101A	Shift	C Shift (22:00–06:00)
Equipment Type	Centrifugal Pump (Main Feed)	Work Order	WO-2026-204417
Manufacturer	Grundfos Industries	Model	NB-100-250/220
Installed On	14-Feb-2019	Serial No.	GFI-88712-VZK
Inspector	M. Suresh (Sr. Instrument Tech)	Emp ID	EMP-10234
Supervisor	P. Lakshmi, Plant Reliability Eng.	Approval Date	23-Aug-2026

1. Equipment Overview & History

Pump PMP-CDU4-P101A is the primary crude feed pump serving the No. 4 Crude Distillation Unit (CDU-4) at Visakhapatnam Refinery. The pump handles crude oil at 65 degrees C with a rated flow of 320 m3/h at 7.2 bar differential pressure. It has logged 31,200 operating hours since commissioning. The mechanical seal assembly (Model: John Crane Type 8B1) was last replaced in January 2025 (WO-2025-119042) following a seal face leak that caused minor product loss. Vibration trending on the drive-end bearing has shown a gradual upward trend over the last two months (from 2.1 to 3.8 mm/s RMS). A cavitation event was reported by operator S. Patel on 15-Aug-2026 during the evening shift, lasting approximately 12 minutes before suction conditions stabilised.

2. Inspection Findings — Sensor & Parameter Readings

Parameter / Sensor	Tag ID	Measured	Normal Range	Status
Vibration - Drive End Bearing	VT-P101A-DE	3.8 mm/s RMS	≤ 3.5	Marginal
Vibration - Non-Drive End Bearing	VT-P101A-NDE	1.9 mm/s RMS	≤ 3.5	Normal
Bearing Temperature - DE	TT-P101A-BRG1	72 °C	≤ 80 °C	Normal
Suction Pressure	PT-P101A-SUC	1.82 bar	1.8 – 2.2 bar	Normal
Discharge Pressure	PT-P101A-DIS	9.06 bar	8.9 – 9.5 bar	Normal
Seal Chamber Pressure	PT-P101A-SL	4.4 bar	4.0 – 5.0 bar	Normal
Motor Current Draw	CT-P101A-MOT	48.2 A	42 – 52 A	Normal
Flow Rate	FT-CDU4-FEED	298 m3/h	290 – 330 m3/h	Normal
Noise Level (1 m distance)	—	86 dBA	≤ 82 dBA	Above Limit
Lube Oil Pressure	PT-P101A-OIL	2.1 bar	2.0 – 2.5 bar	Normal

3. Component-Level Inspection

Component	Condition	Action Taken
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Mechanical Seal (John Crane 8B1)	Satisfactory — no leakage detected	None
Drive End Bearing (SKF 6314-2RS)	Marginal vibration — trending upward	Grease topped up; monitor weekly
Non-Drive End Bearing (SKF 6311)	Normal	None
Impeller	Not opened — scheduled borescope pending	WO raised: WO-2026-205001
Suction Strainer	Partially blocked — 18% open area loss	Cleaned in-situ; pressure drop OK
Coupling (Rexnord Thomas Disc)	Good alignment — 0.04 mm offset	None
Gland Packing / Seal Housing	Normal — slight moisture on gland, acceptable	Monitored; no action
Motor (ABB M3BP-225SM)	Normal — no abnormal heat detected	None
Baseplate Anchor Bolts	All bolts torqued to spec	None

4. Lubrication & Materials Used

Bearing lubrication topped up with Mobil SHC Polyrex EM (NLGI Grade 2 polyurea grease). Quantity: 35 grams per bearing cavity. Seal flush fluid is clean condensate (conductivity < 5 microS/cm) supplied from the CDU-4 overhead condensate drum. No process oil detected in seal fluid sample taken on this date.

5. Root Cause — Cavitation Event (15-Aug-2026)

Investigation by technician M. Suresh and process engineer V. Reddy concluded that the cavitation event on 15-Aug-2026 was caused by a temporary drop in suction header pressure below 1.7 bar due to the concurrent startup of PMP-CDU4-P102B (the standby pump) on the same suction header without adequate sequencing. The event lasted 12 minutes and is believed to have caused minor impeller surface erosion. A borescope inspection is recommended within 30 days to assess impeller condition.

6. Recommended Follow-up Actions

#	Action	Priority	Target Date	Owner
1	Borescope impeller inspection (WO-2026-205001)	High	15-Sep-2026	M. Suresh / Reliability
2	Repeat vibration trending — DE Bearing weekly	Medium	Ongoing	Condition Monitoring
3	Review pump startup sequencing procedure SOP-PMP-10044	Medium	30-Aug-2026	V. Reddy (Process)
4	Replace DE Bearing if vibration exceeds 4.5 mm/s	Conditional	On trigger	Maintenance
5	Update CMMS equipment history with this inspection	Routine	25-Aug-2026	M. Suresh

7. Cross-References

Record Type	System	Reference ID
P&ID Drawing	Eng. Document Vault	PID-CDU4-FEED-007, Rev. D
Prior Work Order (seal replacement)	CMMS (SAP PM)	WO-2025-119042
Cavitation Event Report	Shift Handover App	EVT-2026-0815-CDU4
OEM Pump Manual	Doc Management	OEM-GFI-NB100250-R3
LOTO Procedure	SOP Repository	SOP-SAF-0021

SYNTHETIC DOCUMENT — For Knowledge Graph testing purposes only. All names, IDs, and figures are fictitious.